

William Nicholas McWilliams

Department of Agricultural and Applied Economics
Virginia Polytechnic Institute and State University

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EDUCATION

- **Virginia Polytechnic Institute and State University** Blacksburg, Virginia
Ph.D. in Agricultural and Applied Economics August 2023 – Expected Summer 2027
 - Dissertation: “Measuring, Monitoring, and Forecasting Across the Agri-Food Value Chain”
 - Committee: Dr. Olga Isengildina Massa (Co-Chair), Dr. Shamar Stewart (Co-Chair), Dr. Clint Neill, Dr. Mario Orteiz, Dr. David Kohl
- **Louisiana Tech University** Ruston, Louisiana
B.S. in Finance May 2018

RESEARCH INTERESTS

Agricultural Finance, Commodity Markets, Agrifood Supply Chains, Market Structure and Competition, Intersection of Theory-Driven and Algorithmic Approaches to Forecasting and Inference

PUBLICATIONS

- McWilliams, W. N., S. Stewart, and O. Isengildina Massa. “Food Price Inflation Forecasting: Insights from a Macroeconomic Auto-Regressive Random Forest Approach.” *Food Policy* 142 (2026): 103115.
<https://doi.org/10.1016/j.foodpol.2026.103115>

WORKING PAPERS

- “Measuring Agricultural Financial Conditions: A Mixed-Frequency Dynamic Factor Approach,” with Olga Isengildina Massa and Shamar Stewart.

CONFERENCE PRESENTATIONS

- “Measuring Agricultural Financial Conditions: A Mixed-Frequency Dynamic Factor Approach.” Agricultural and Applied Economics Association Annual Meeting, Kansas City, MO, 2026 (poster)
- “Food Price Inflation Forecasting: Insights from a Macroeconomic Auto-Regressive Random Forest Approach.” NCCC-134 Conference on Applied Commodity Price Analysis, Forecasting, and Market Risk Management, 2025
- “Food Price Inflation Forecasting: Insights from a Macroeconomic Auto-Regressive Random Forest Approach.” Agricultural and Applied Economics Association Annual Meeting, 2024

TEACHING EXPERIENCE

- **Virginia Polytechnic Institute and State University** Blacksburg, Virginia
Instructor of Record Summer 2025 – Summer 2026
 - AAEC 2104: Personal Financial Planning (Summer 2026) – 75 students, online asynchronous delivery
 - AAEC 4504: Agricultural Price and Market Analysis (Spring 2026) – 28 students
 - AAEC 2104: Personal Financial Planning (Summer 2025) – 103 students, online asynchronous delivery
- **Virginia Polytechnic Institute and State University** Blacksburg, Virginia
Graduate Teaching Assistant Fall 2024 – Spring 2025
 - AAEC 4504: Agricultural Price and Market Analysis (Dr. Olga Isengildina Massa)
 - AAEC 4804/STAT 4804/AAEC 5804G: Fundamentals of Econometrics (Dr. Shamar Stewart)
 - AAEC 3514: Agricultural Futures and Options (Dr. Olga Isengildina Massa)
- **Louisiana Tech University** Ruston, Louisiana
Graduate Teaching Assistant August 2022 – May 2023
 - Finance 318 (Business Finance), Economics 510 (Managerial Economics), Finance 319 (Principles of Finance)

HONORS, AWARDS, AND FELLOWSHIPS

- **COINS Graduate Fellow:** Lead Wheat Analyst, Commodity Investing by Students (COINS) – Student-managed organization investing \$1.3+ million of the university’s endowment in commodity markets (Fall 2024 – Present)
- **USDA National Needs Fellowship:** National Institute of Food and Agriculture – Fellowship for outstanding students with expertise in commodity markets (Fall 2023 – Present)
- **Farm Foundation Agricultural Scholar:** Year-long program for graduate students in agricultural economics and policy (2024)

GRANTS

- **USDA National Institute of Food and Agriculture, AFRI Predoctoral Fellowship (A7101):** “Measuring Agricultural Financial Conditions: A Mixed-Frequency Dynamic Factor Approach.” Project Director. \$180,000 requested over three years. Submitted April 2026. *Under review*

PROFESSIONAL SERVICE

- **Graduate Student Representative:** Graduate Committee, Department of Agricultural and Applied Economics, Virginia Polytechnic Institute and State University (Fall 2026 – Present)
- **Journal Referee:** Food Policy
- **Presentations Reviewer:** Agricultural and Applied Economics Association Annual Conference, First Round Reviewer, Econometrics Section (2025)

RESEARCH EXPERIENCE

- **Virginia Polytechnic Institute and State University** Blacksburg, Virginia
Graduate Research Assistant August 2023 – Present
 - Supervisors: Dr. Olga Isengildina Massa and Dr. Shamar Stewart
 - Conduct original research in commodity price analysis, forecasting, and market risk management
 - Develop and apply advanced time-series econometric models and machine learning algorithms to analyze agrifood systems
 - Investigate price discovery mechanisms, market integration, and volatility spillovers in agricultural futures markets

PROFESSIONAL EXPERIENCE

- **CME Group** Chicago, Illinois
Intern, Agricultural Research and New Product Development June – August 2026
 - Conducted research on new and existing agricultural derivatives products, working across the research and business lines
 - Produced supporting research and written materials informing decisions on contract specifications and product design
- **Catholic Relief Services** Dili, Timor-Leste
Organizational Development Volunteer August – December 2020
 - Facilitated strategic planning for Lanamona Credit Union (11 staff, 1,436 members)
 - Conducted risk assessment and developed mitigation strategies for microfinance operations
- **Peace Corps** Dili, Timor-Leste
Community Economic Development Volunteer August 2017 – May 2018
 - Developed strategic plans for cooperative development, addressing market systems development needs and value chain inequities for vanilla and coffee farmers
 - Designed and delivered trainings on cooperative development, processing techniques, and international market access for vanilla and spice cultivation
 - Provided management and operations consulting for RADIFU, Municipio Ainaro’s largest savings and loans cooperative
- **Graphic Packaging International** Monroe, Louisiana
Financial Analyst Intern May – August 2017
 - Analyzed plant financial statements to identify revenue growth drivers
 - Conducted project costing and financial closure processes

TECHNICAL SKILLS

- **Statistical Software:** R, MATLAB, Python, Excel
- **Languages:** English (native), Tetun (Advanced)

PROFESSIONAL AFFILIATIONS

Agricultural and Applied Economics Association (AAEA)

REFERENCES

- **Dr. Olga Isengildina Massa**

- Professor of Agricultural and Applied Economics, Virginia Polytechnic Institute and State University
- 321-A Hutcheson Hall, Blacksburg, VA 24061
- Email: oimassa@vt.edu

- **Dr. Shamar Stewart**

- Assistant Professor of Agricultural and Applied Economics, Virginia Polytechnic Institute and State University
- 202-B Hutcheson Hall, Blacksburg, VA 24061
- Email: stewartls@vt.edu

- **Dr. William R. McCumber**

- Associate Dean of Graduate Programs, College of Business, Louisiana Tech University
- Ruston, LA 71272
- Email: mccumber@latech.edu